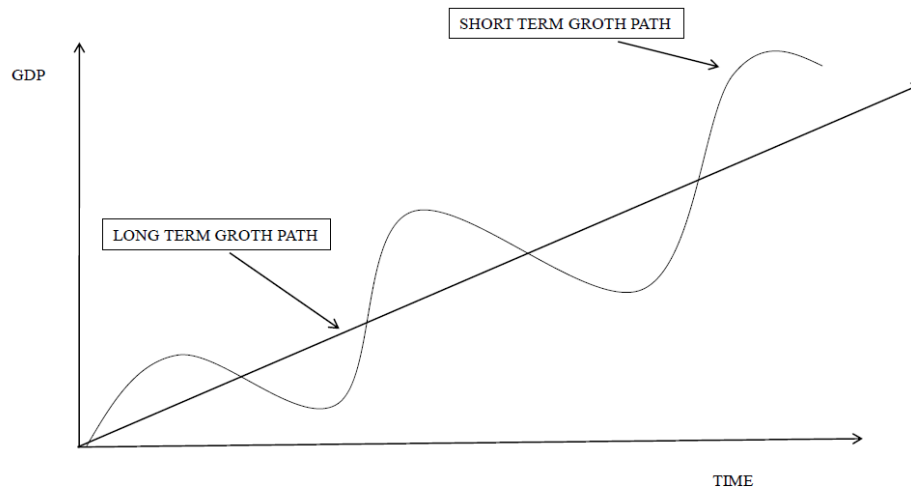


Nature of an Economy (Business Cycle)



1. Long-Term Growth Path

- Shown as a straight, upward-sloping line.
- Represents the **average or potential growth** of the economy over time.
- Indicates the trend of increasing productive capacity due to factors like technological progress, capital accumulation, and labor force growth.
- This is sometimes called the "trend line" of economic growth.

2. Short-Term Growth Path (Business Cycle)

- Shown as a wavy line oscillating above and below the long-term growth path.
- Represents the **actual GDP**, which fluctuates due to short-term economic factors.
- These fluctuations are known as the **business cycle**.

Phases of the Business Cycle

The short-term growth path (wavy line) shows the economy moving through different phases:

- **Expansion:** GDP rises above the long-term trend; economic activity is strong.
- **Peak:** The highest point of the cycle; growth slows down.
- **Contraction (Recession):** GDP falls below the long-term trend; economic activity declines.
- **Trough:** The lowest point of the cycle; the economy begins to recover.

The business cycle refers to the natural rise and fall of economic activity over time, characterized by fluctuations in real GDP, employment, and other economic indicators. It consists of four main phases:

1. Expansion

- This phase marks a period of increasing economic activity.
- Key indicators such as employment, income, production, and consumer spending rise.
- Consumer confidence improves, leading to more spending and investment.
- Businesses grow and profits typically increase

2. Peak

- The peak is the highest point of the business cycle, where economic activity reaches its maximum.
- At this stage, growth slows down and economic indicators level off.
- Prices tend to be at their highest, and the economy may "overheat," meaning demand outstrips supply.

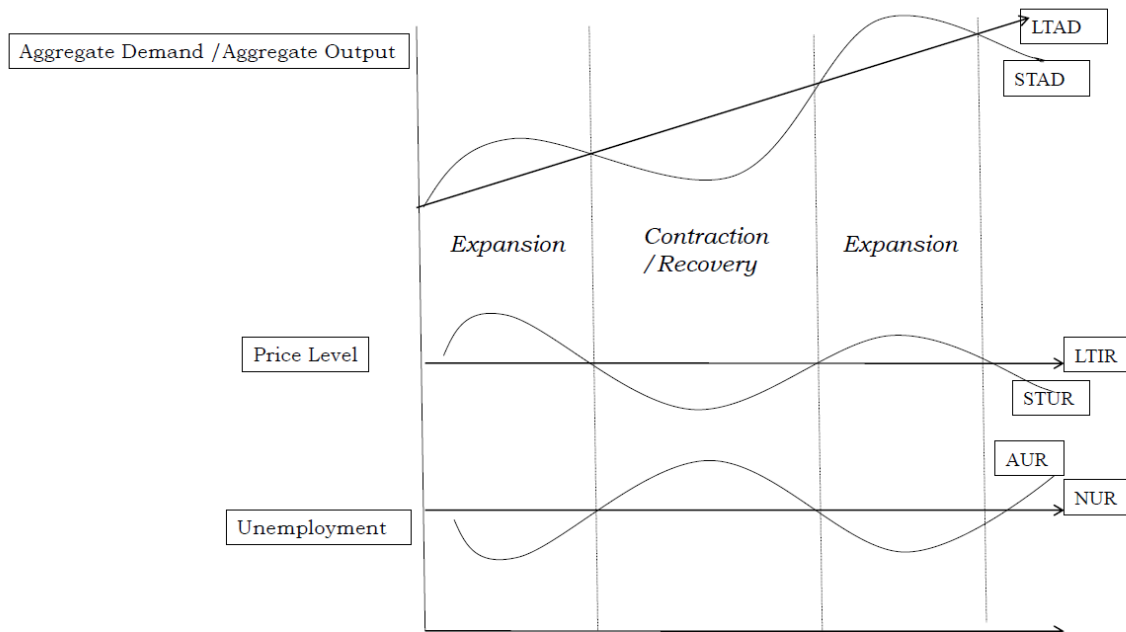
3. Contraction (Recession)

- Following the peak, the economy enters a contraction phase.
- Economic activity declines: production, income, and employment fall.
- Consumer spending decreases, and businesses may reduce output and lay off workers.
- This phase continues until the economy reaches its lowest point.

4. Trough

- The trough is the lowest point of the cycle, marking the end of contraction.
- Economic activity stabilizes and begins to recover.
- This phase leads into the next expansion, restarting the cycle.

Business Cycle and Relationships among Growth, inflation and unemployment



The diagram presents three main variables over time:

- Aggregate Demand/Aggregate Output (Growth)
- Price Level (Inflation)
- Unemployment

Each variable is shown as a wavy line (short-term fluctuations) around a straighter line (long-term trend).

1. Growth (Aggregate Output)

- Wavy Line (STAD): Short-term changes in output due to the business cycle.
- Upward Line (LTAD): Long-term growth trend.
- Expansion: Output rises above the long-term trend.
- Contraction/Recovery: Output falls below the trend, then recovers.

2. Inflation (Price Level)

- Wavy Line (STUR): Short-term inflation rate fluctuates with the cycle.
- Upward Line (LTIR): Long-term inflation trend.
- Expansion: Inflation rises above the long-term trend.
- Contraction: Inflation falls below the trend, possibly leading to disinflation.

3. Unemployment

- Wavy Line (AUR): Actual unemployment rate fluctuates with the cycle.
- Flat Line (NUR): Natural unemployment rate (long-term average).
- Expansion: Unemployment falls below the natural rate.
- Contraction: Unemployment rises above the natural rate.

Relationships among growth, Inflation and Unemployment

During Expansion-

- Growth: Output rises above trend.
- Inflation: Price level increases faster than average.
- Unemployment: Drops below the natural rate.

Relationship-

When the economy grows quickly (expansion), inflation tends to rise and unemployment falls.

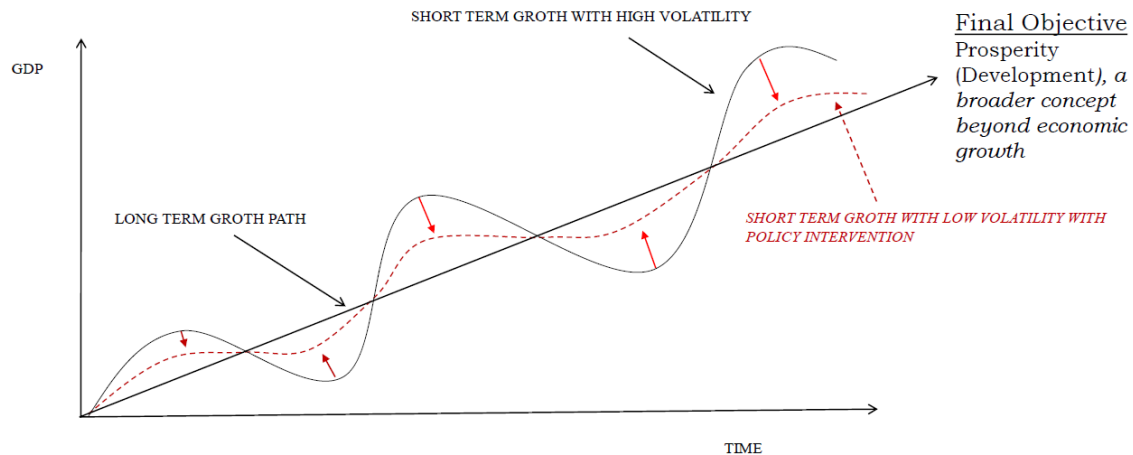
During Contraction/Recovery

- Growth: Output falls below trend.
- Inflation: Price level rises more slowly or may fall.
- Unemployment: Rises above the natural rate.

Relationship-

When the economy slows down (contraction), inflation slows or may turn negative, and unemployment rises.

Objectives of Macroeconomic Management (Stability)



Long-Term Growth Path

- The straight, upward-sloping line.
- Shows the ideal or potential growth of the economy if it expands smoothly, without major disruptions. This is often called the “trend growth rate.”
- Reflects improvements in productivity, technology, labor force, and capital over time.

Short-Term Growth with High Volatility

- The wavy black line that moves above and below the long-term growth path.
- Illustrates real-world economic cycles—periods of rapid growth (booms) followed by slowdowns or recessions (busts).
- High volatility can lead to economic instability, uncertainty, higher unemployment during downturns, and inflation during booms.

4. Short-Term Growth with Low Volatility (Policy Intervention)

- The smoother, dashed red line.
- Represents the effect of effective macroeconomic policies (like monetary and fiscal policy), which aim to reduce the amplitude of economic cycles.
- To keep actual GDP closer to the long-term growth path, minimizing the negative effects of booms and busts.

5. Prosperity (Development)

- Highlighted at the end of the diagram.
- The ultimate goal of macroeconomic management is not just economic growth, but prosperity—a broader concept that includes stability, reduced poverty, higher living standards, and overall well-being.

Why Is Stability Important?

- **Reduces Uncertainty** - Stable economies encourage investment, business planning, and consumer confidence.
- **Prevents Extreme Outcomes** - Avoids deep recessions (which cause unemployment and hardship) and unsustainable booms (which can lead to inflation and financial crises).
- **Promotes Sustainable Growth** - Stability allows for steady improvements in living standards and reduces the risk of long-term damage from economic shocks.

How Do Policymakers Achieve Stability?

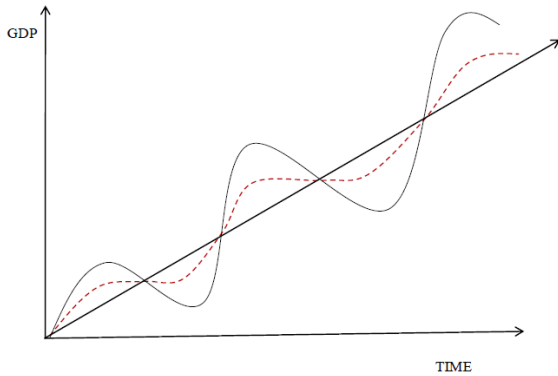
1. **Monetary Policy** - Central banks adjust interest rates and control money supply to keep inflation low and smooth out business cycles.

2. **Fiscal Policy** - Governments use spending and taxation to stimulate the economy during downturns and cool it during booms.

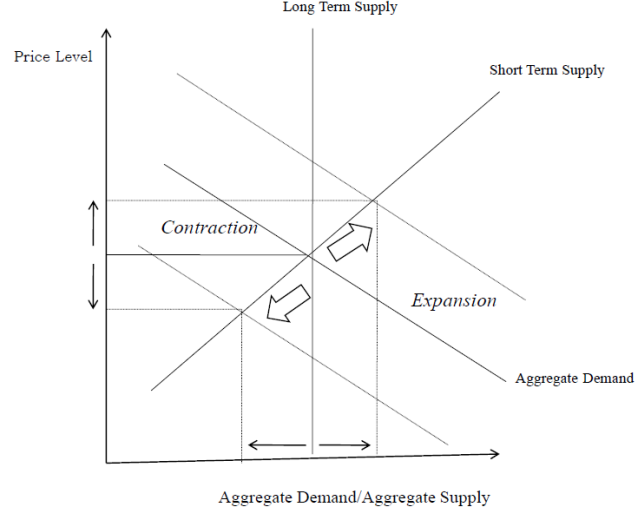
3. **Regulation and Supervision** - Ensuring the financial system is stable and resilient to shocks.

Time Dimension of Macroeconomic Activities

Situation over time



Situation at a time



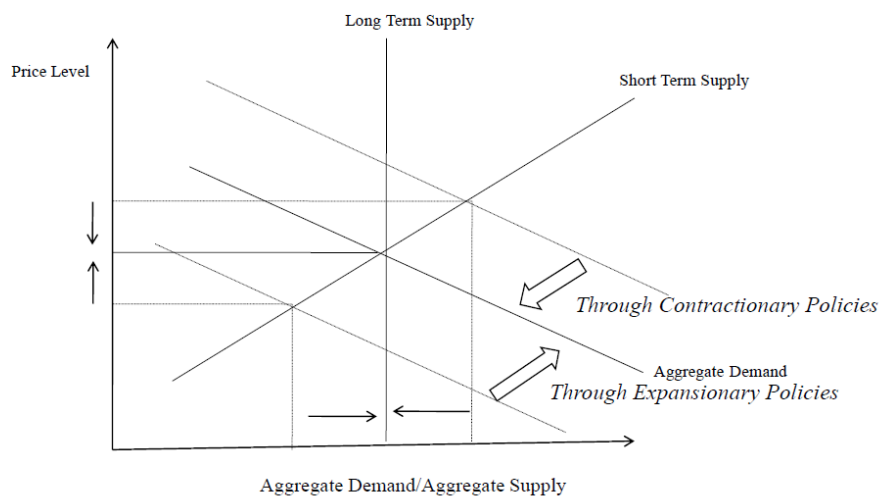
Contraction –

When aggregate demand is low, the equilibrium moves left, leading to lower output and price levels—this is a recession or contraction.

Expansion –

When aggregate demand is high, the equilibrium moves right, leading to higher output and price levels—this is an expansion or boom.

How to Manage Business Cycle to Stabilize the Economy?



This diagram is a classic Aggregate Demand–Aggregate Supply (AD-AS) model used in macroeconomics to illustrate how different policies can stabilize the economy.

- Long Term Supply (Vertical Line)- Represents the economy's potential output (full employment level).
- Short Term Supply (Upward Sloping) - Shows how output responds to changes in demand in the short run.
- Aggregate Demand (Downward Sloping) - Shows the total demand for goods and services at different price levels.

1.Expansionary Policies

Used during recessions or periods of high unemployment to stimulate economic activity. Shift aggregate demand to the right (increase it) to combat recessions.

How it works-

- Aggregate demand is too low (economy below potential output).
- Expansionary policies increase aggregate demand, shifting the AD curve to the right.
- Results in Higher output (GDP) and employment, possibly with a slight increase in the price level (inflation).

Tools-

1.Monetary Policy- Central bank lowers interest rates or increases money supply.

2.Fiscal Policy- Government increases spending and/or cuts taxes.

ex. During the COVID-19 pandemic, many governments increased spending and central banks cut interest rates to support the economy.

2. Contractionary Policies

Used during booms or periods of high inflation to cool down the economy. Shift aggregate demand to the left (decrease it) to combat inflation.

How it works-

- Aggregate demand is too high (economy above potential output, causing inflation).
- Contractionary policies decrease aggregate demand, shifting the AD curve to the left.
- Result is Lower inflation, but may also slow down output and increase unemployment slightly.

Tools-

1.Monetary Policy: Central bank raises interest rates or reduces money supply.

2.Fiscal Policy: Government cuts spending and/or raises taxes.

ex. If inflation rises rapidly, a central bank might increase interest rates to make borrowing more expensive, reducing spending and investment.

Key Macroeconomic Variables

- GDP (both Real and Nominal) and rates of Change
- Price Levels and Rates of Change
- Employment and Unemployment and rates of Change
- Government Budget and Debt
- Balance of Payments (main account and sub accounts)
- Money Supply and rates of Change
- Interest Rates (policy rates and market rates)
- Exchange Rates and their movements